

Zahraa Abdulrazzaq Ziara

Computer Engineering Student

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SUMMARY

Computer Engineering student with a focus on hardware and software integration. Skilled in Assembly, C++, and Embedded Systems with hands-on experience in IoT and Analog Electronics. Proven ability to build functional prototypes and present technical projects in scientific exhibitions. Consistently recognized for high academic performance and a fast-learning pace.

EDUCATION

Bachelor's in Computer Engineering University of Basrah | College of Engineering

Nov 2023 – May 2027 (Expected)

Excellent Academic Standing, Ranked Top 5 in the department.

SKILLS

Technical

- Hardware & Circuits:** Electronics, Electrical Circuits, Digital Logic, ESP32, Arduino.
- Software & Simulation:** Multisim, AutoCAD, MATLAB, Simulink.
- Computing:** C++, HTML/CSS, Assembly (Microprocessors), Operating Systems, Networks, Algorithms.

Professional

- Productivity:** Microsoft Word, PowerPoint.
- Soft Skills:** Technical Problem Solving, Team Collaboration, Fast Learner.

Languages

- Arabic:** Native.
- English:** Fluent.
- Spanish:** Basic.

CERTIFICATIONS AND EXPERIENCE

EXPERIENCE

- Scientific Innovation Exhibition | University of Basrah**

Selected to showcase engineering prototypes and engage with technical faculty and industry representatives.

- Technical University Scientific Fair | Participant**

Demonstrated technical hardware projects and discussed design methodologies with a professional audience.

CERTIFICATIONS

- ENFIELD AI Summer School (2025):** Intensive training on Adaptive AI, Green AI, and Trustworthy AI systems.
- Web Design & Development:** Training in HTML/CSS for responsive interface design.
- Network Engineering:** Proficient in networking protocols, routing, and switching.

PROJECTS

IoT Surveillance Vehicle

- **Technologies:** ESP32 Microcontroller, C++, WebSockets, Embedded Web Server.
- Designed and implemented a remotely operated surveillance vehicle featuring real-time video streaming to a custom-built mobile web interface.
- Integrated an ESP32-CAM module to handle concurrent motor control and low-latency video transmission via local Wi-Fi.
- Developed a responsive web-based dashboard for directional control and visual monitoring.

Active Frequency Spectrum Analyzer

- **Technologies:** Operational Amplifiers (Op-Amps), Active Filters, Integrated Circuits (ICs), Analog Signal Processing.
- Engineered a hardware-only audio visualizer that processes live sound signals into frequency bands without the use of a microcontroller.
- Designed Active Low-Pass (LPF) using Op-Amps to isolate specific audio frequencies.
- Implemented a multi-stage amplification and LED-driver circuit to provide real-time visual feedback based on acoustic input.